

THATOE NYI

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PROFESSIONAL SUMMARY

Final-year Computer Science student at Rangsit University (expected May 2027) who builds and deploys applied AI systems end to end: data collection, labelling, training, evaluation and serving behind an API. Hands-on with computer vision (YOLOv8 object detection, Facenet512 face embeddings, OpenCV, OCR), machine learning for anomaly detection (scikit-learn), and LLM/RAG pipelines deployed with Docker. An earlier start in petroleum engineering gives a feel for industrial processes; the goal now is AI that makes manufacturing lines faster and more reliable. Available for a full-time, on-site internship from mid-December 2026.

TECHNICAL SKILLS

- **Programming:** Python (primary), TypeScript, SQL, Dart
- **ML & Deep Learning:** PyTorch (Ultralytics YOLOv8), TensorFlow/Keras (DeepFace), scikit-learn (K-Means, DBSCAN, Isolation Forest), neural networks, decision trees, pandas, NumPy, Matplotlib, Seaborn
- **Computer Vision & Image Processing:** object detection, image augmentation, Roboflow annotation, face embeddings (Facenet512), OpenCV, Tesseract OCR, QR decoding (pyzbar)
- **LLM & RAG:** LangGraph, LiteLLM, Gemini API, pgvector, RAG evaluation (LLM-as-judge)
- **MLOps & Deployment:** FastAPI, Flask, Docker, docker-compose, Linux VPS, Google Colab (GPU training), PostgreSQL, Supabase
- **Automation & Engineering Practice:** n8n, PyAutoGUI, OpenPyXL, web scraping, REST APIs/webhooks; Git/GitHub, pytest, SDLC from spec to deployment, Claude Code, GitHub Copilot

SELECTED PROJECTS

Snap & Go — Vision-Based Self-Checkout | [GitHub](#) — *Python, PyTorch/YOLOv8, OpenCV, FastAPI, Roboflow, Supabase, Next.js*

- Built a custom object-detection dataset from scratch: recorded product videos, extracted ~100 frames per class and labelled bounding boxes in Roboflow.
- Applied brightness ($\pm 25\%$), blur and rotation ($\pm 15^\circ$) augmentation for robustness to fluorescent lighting; trained YOLOv8n for 50 epochs on a Colab T4 GPU.
- Deployed behind FastAPI with OpenCV decoding, a configurable confidence threshold (0.80) and product/price lookup; added QR (pyzbar) payment-slip verification and pytest checks.

Face Recognition System | [GitHub](#) — *Python, TensorFlow/Keras (DeepFace), OpenCV, Flask*

- Built a 105-identity recognizer by extracting 512-dimensional Facenet512 embeddings into a reference database, avoiding costly model training.
- Matched by cosine distance with a tuned threshold (0.4), served top-5 matches via a Flask REST API, and documented model and detector trade-offs (VGG-Face/ArcFace, MTCNN/RetinaFace).

Unsupervised Anomaly Detection — Bank Transactions | [GitHub](#) — *Python, scikit-learn, pandas, Seaborn*

- Analysed 2,512 unlabelled transactions; log-transformed skewed amounts (skew 1.74) and engineered behavioural features (per-account averages, amount-to-balance, accounts per device).
- Compared K-Means distance-to-centroid (126 flags), DBSCAN density outliers (17 flags) and Isolation Forest (126 flags), with cluster and outlier visualisations.

Agentic Content Pipeline (Jarvis) | [GitHub](#) — *Python, FastAPI, LiteLLM, Gemini, PostgreSQL/pgvector, Docker*

- Multi-agent pipeline turning a voice/text brief into a bilingual post and image, grounded in a pgvector knowledge base, with human-in-the-loop approval; deployed with Docker on a VPS.

RAG Customer-Support Agent with Human Handoff | [GitHub](#) — *Python, FastAPI, Gemini, custom RAG, Telegram/Messenger APIs*

- Built for a real client: answers from client documents via embedding search, escalates to a human on Telegram, and includes an admin app for no-code knowledge updates.

RESEARCH EXPERIENCE

Senior Research Project: Chunking Quality vs. RAG Architecture — *Rangsit University (starts Jan 2027)*

- Designing a controlled experiment on whether chunk-quality metrics predict answer quality across naive, Self-RAG and agentic retrieval; proposal chapters 1–3 drafted.

Assistant Researcher, AI Literacy Study — *Rangsit University*: built the Next.js/Supabase learning platform used to run and log a quasi-experimental study.

EDUCATION

Rangsit University, Pathum Thani, Thailand — B.S. Computer Science, expected May 2027

Relevant coursework: Computer Vision, Data Structures & Algorithms, Agentic AI / RAG Systems, Process Automation (RPA), Mobile Programming

Yangon Technological University, Yangon, Myanmar — B.E. Petroleum Engineering, 2015–2019 (transferred to CS)

CERTIFICATIONS

- **Machine Learning Specialization** — DeepLearning.AI & Stanford Online, completed Oct 2026
- **Meta Data Analyst Professional Certificate · IBM RAG and Agentic AI Professional Certificate** (in progress)

ADDITIONAL EXPERIENCE

Educator & Private Tutor — 2015–2024: taught GED/IGCSE mathematics and analytical reasoning one-to-one.

Languages: Burmese (native), English (professional)